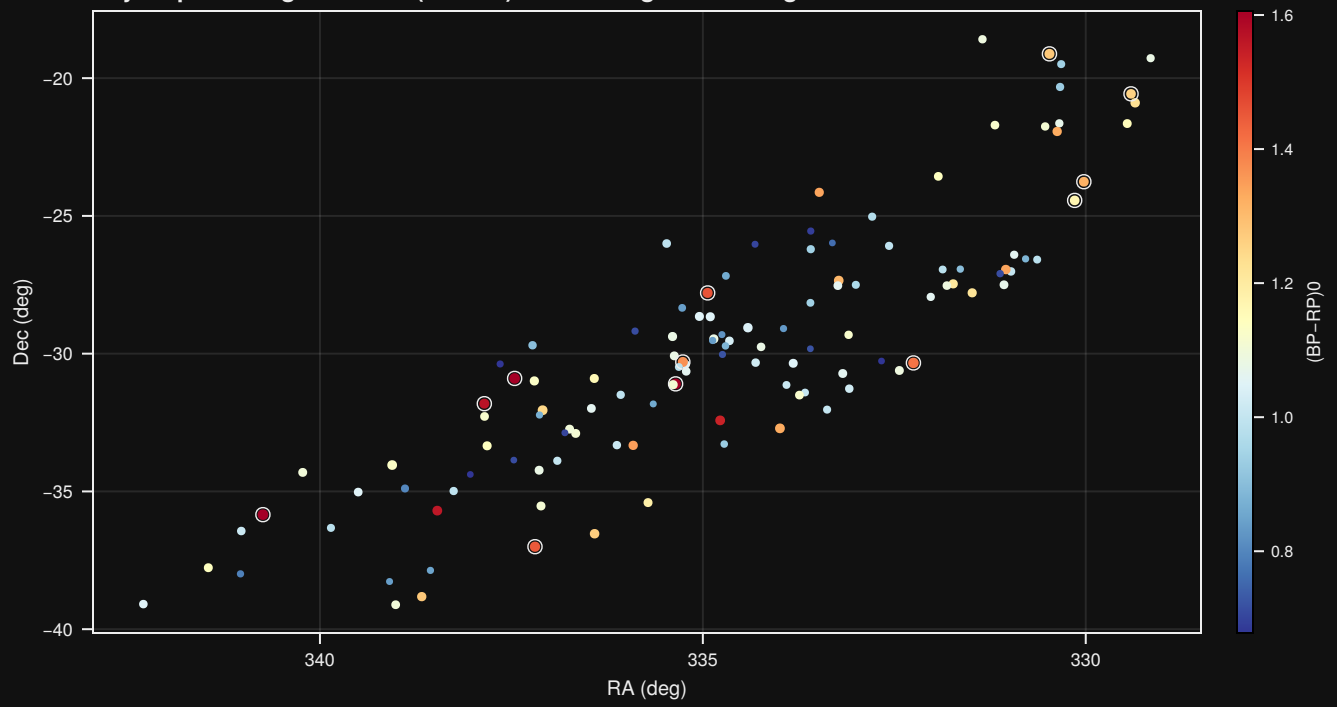
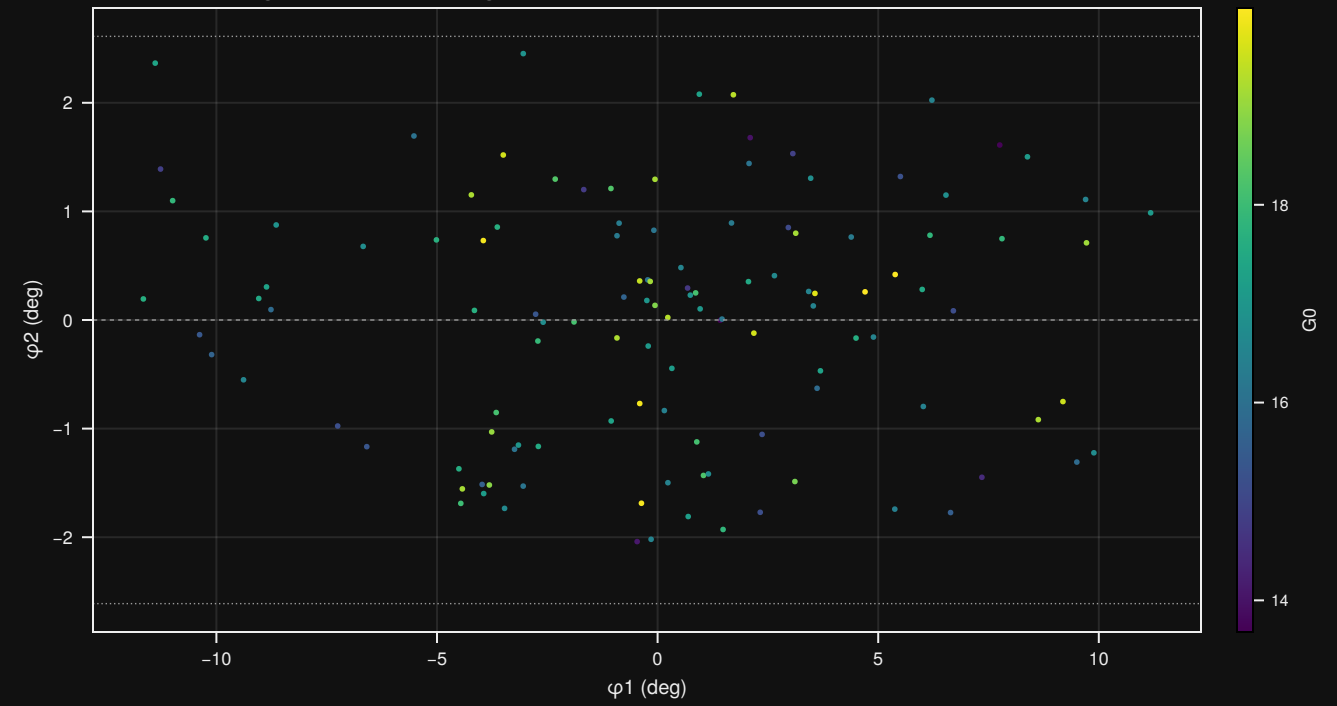
New-25

MEMBERSHIP	
Stars in catalog	121
with Gaia GSP-Phot	96 (79%)
with radial velocity	12
Stream id (sID)	81
GEOMETRY · great-circle fit	
Centroid RA, Dec	334.7, -30.3 deg
Centroid l, b	18.6, -56.5 deg
Pole RA, Dec	50.2, 23.4 deg
Length (φ1, 2–98%)	20.9 deg
Width σ(φ2), robust	1.31 deg
KINEMATICS	
pmRA* med ± σ	0.82 ± 0.36 mas/yr
pmDE med ± σ	-2.45 ± 0.50 mas/yr
μφ1* med ± σ	2.60 ± 0.62 mas/yr
μφ2 med ± σ	-0.38 ± 0.22 mas/yr
Parallax (cat.) med	0.033 ± 0.071 mas
Parallax (wtd, ZP-corr.)	0.047 ± 0.011 mas
Distance	d = 21.1 ± 5.0 kpc
Physical length	7.69 kpc
RV median ± σ(MAD)	-47.1 ± 63.0 km/s
RV gradient dV/dφ1	5.25 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.7–20.0 (17.0)
(BP−RP)0 median	1.06
GSP-Phot [M/H] (n=96)	-0.88 ± 0.96
GSP-Phot Teff median	5095 K
RUWE > 1.4	0.0 %
PM clusters (largest)	6 (14 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	121 | 121
n_RV catalog | paper	12 | 12
<[Fe/H]> paper	-1.68
Stellar mass (paper)	18000 Msun
Pole offset vs paper	0.0 deg
Remark	Velocities unclear, superposed

Sky map · catalog members (n = 121) · size ∝ brightness · ring = has RV

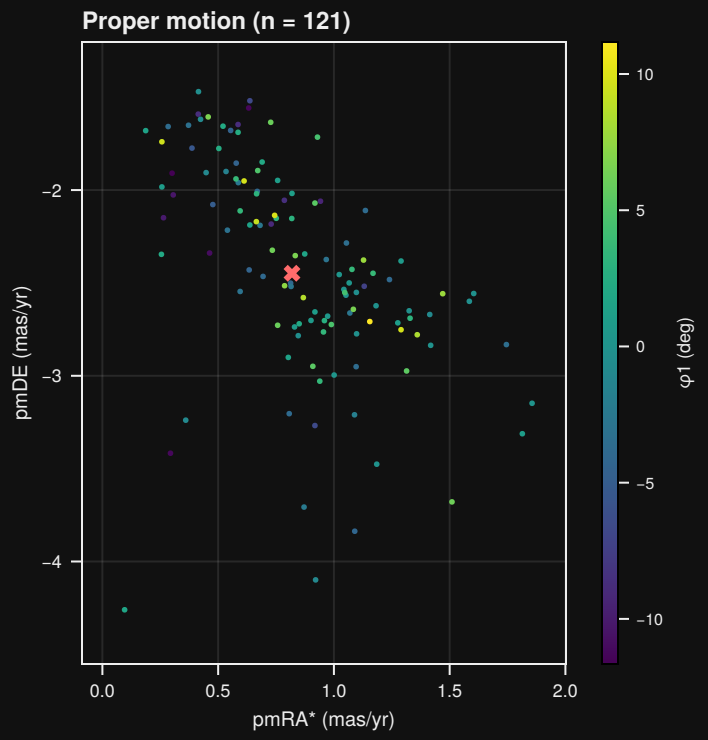


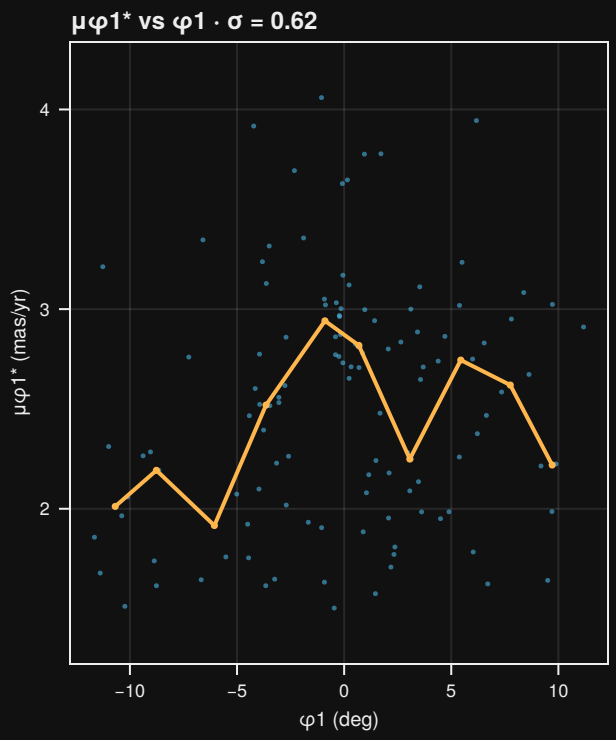
Stream frame (great-circle fit) · length 20.9° · width σ 1.31°

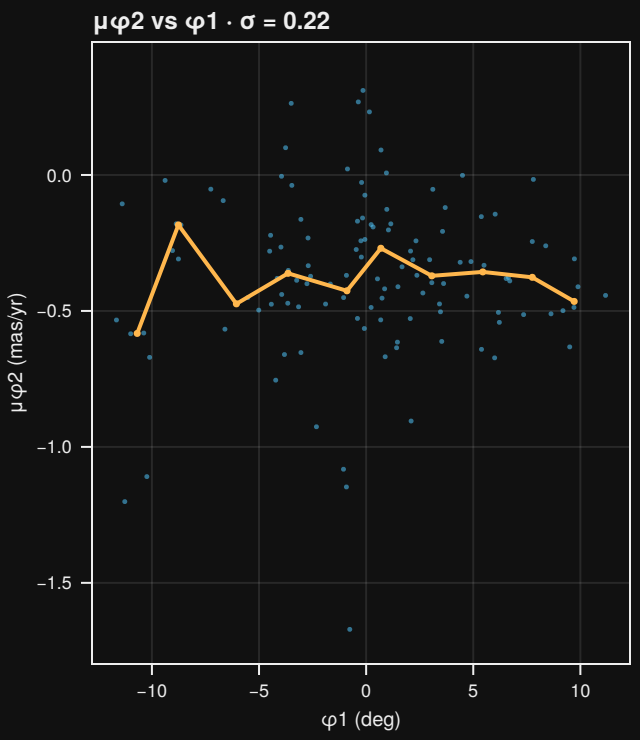


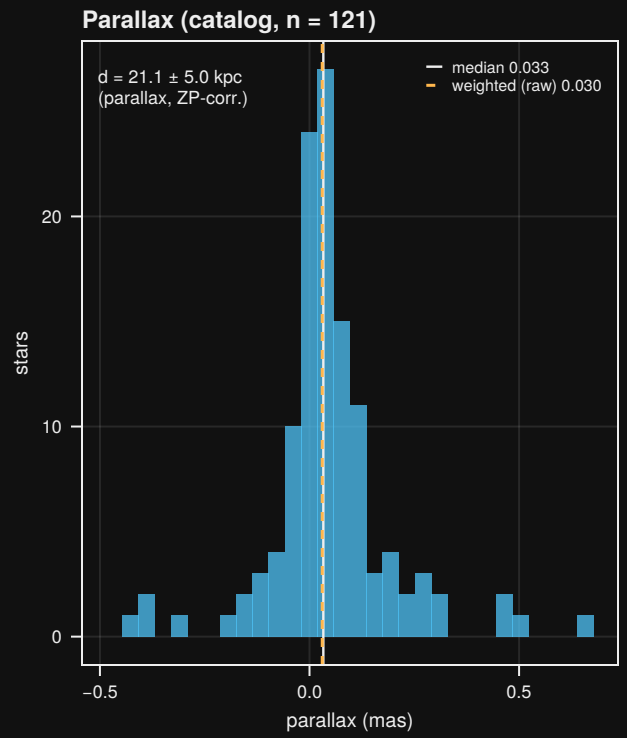
New-25

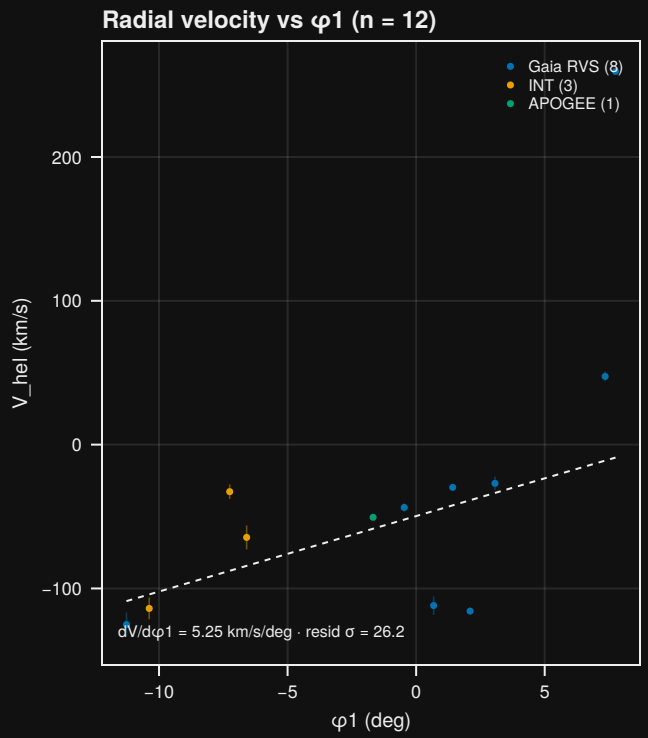
Kinematics · proper motion · parallax · radial velocity

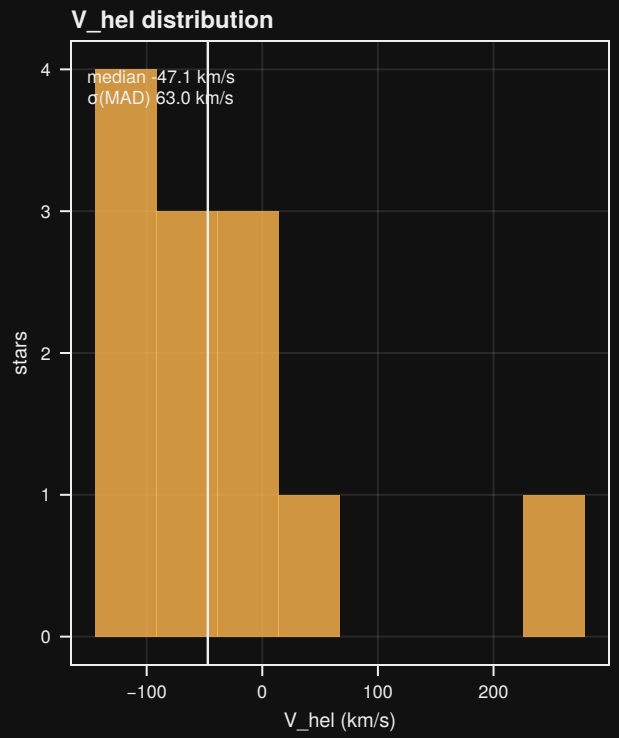






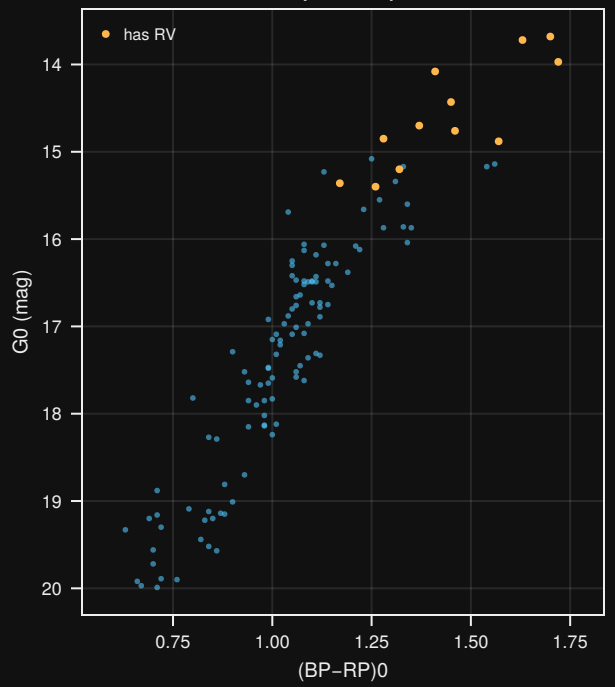




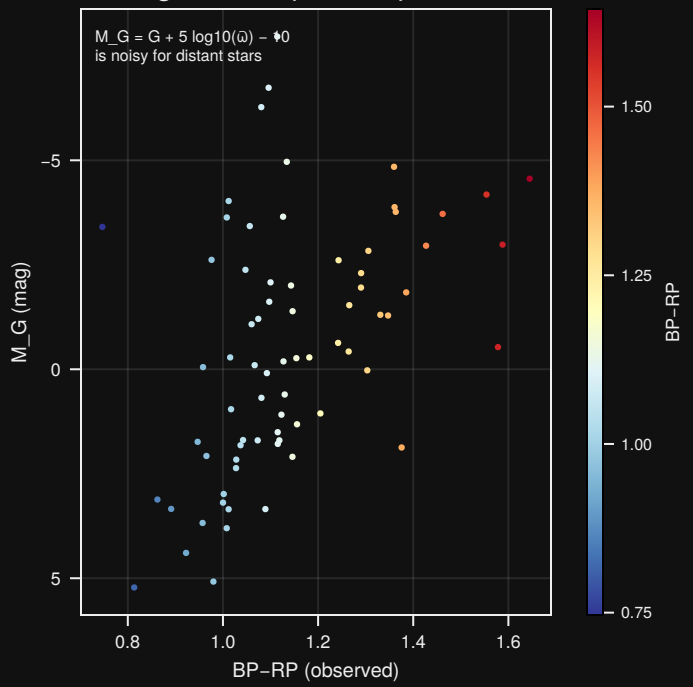


New-25

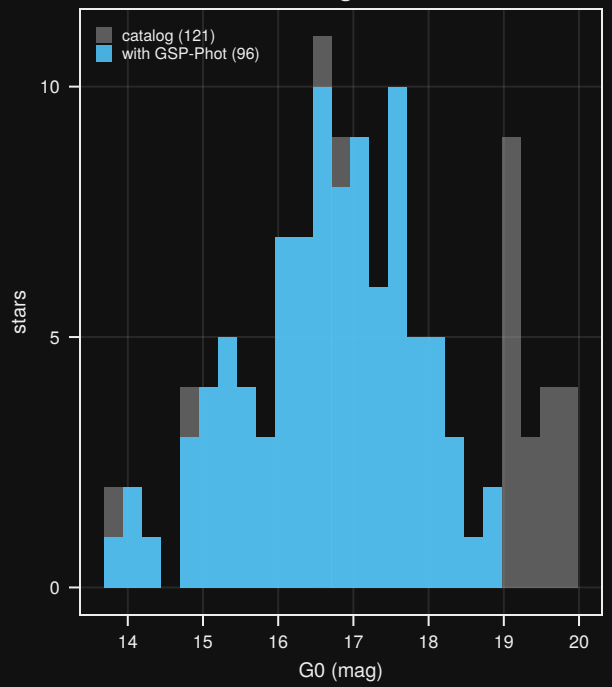
CMD · dereddened (n = 121)



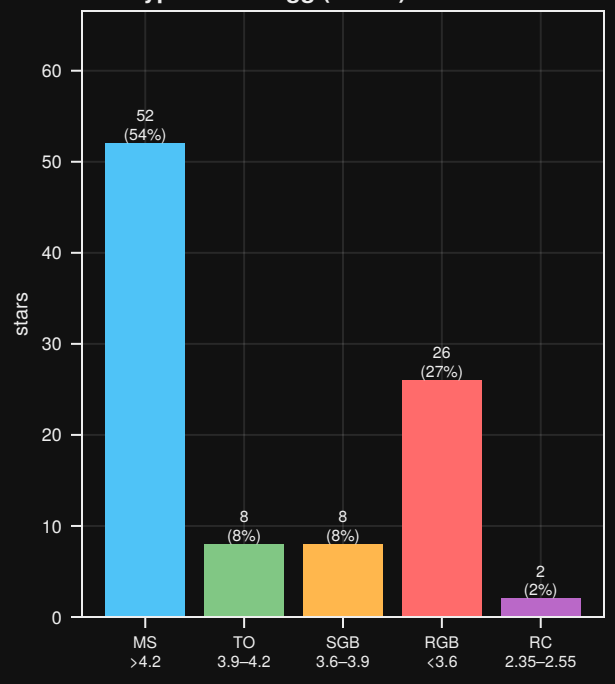
HR diagram · ϖ>0 (n = 73/96)



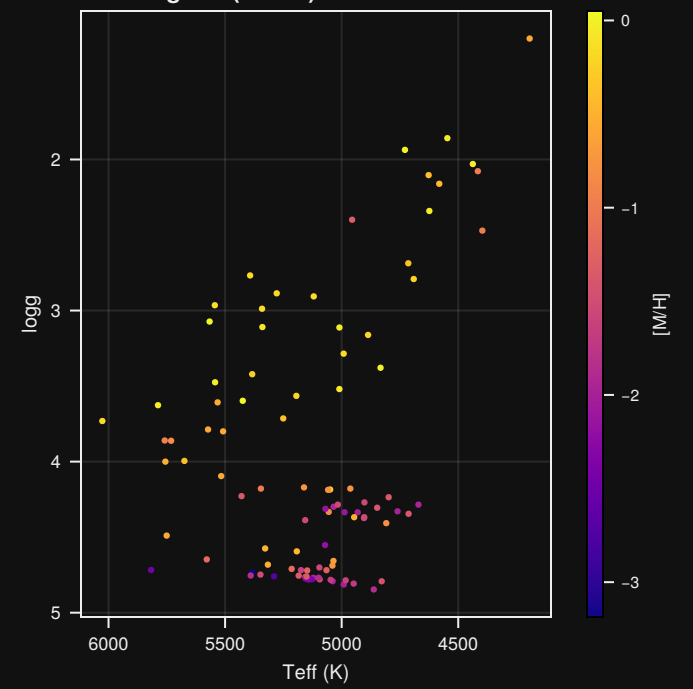
G0 & GSP-Phot coverage



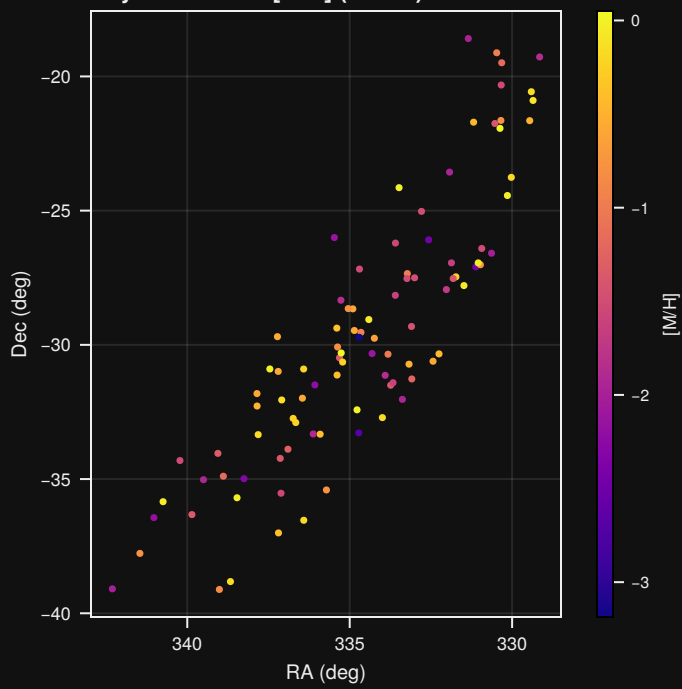
Star types from logg (n = 96)



Kiel diagram (n = 96)



Sky · GSP-Phot [M/H] (n = 96)

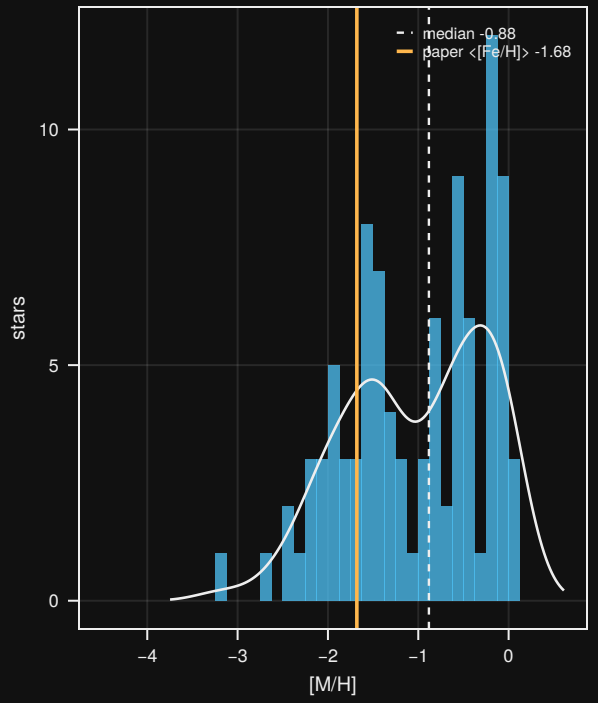


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

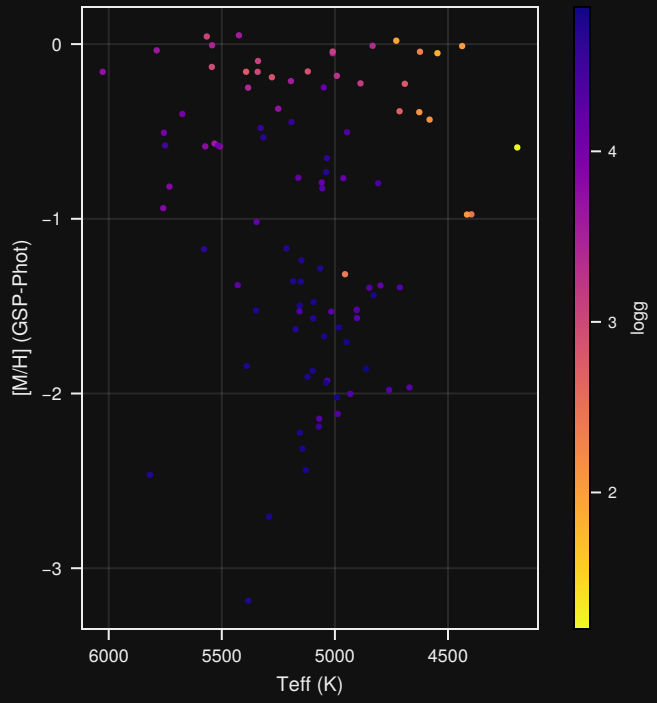
New-25

Metallicity · age · astrometric quality

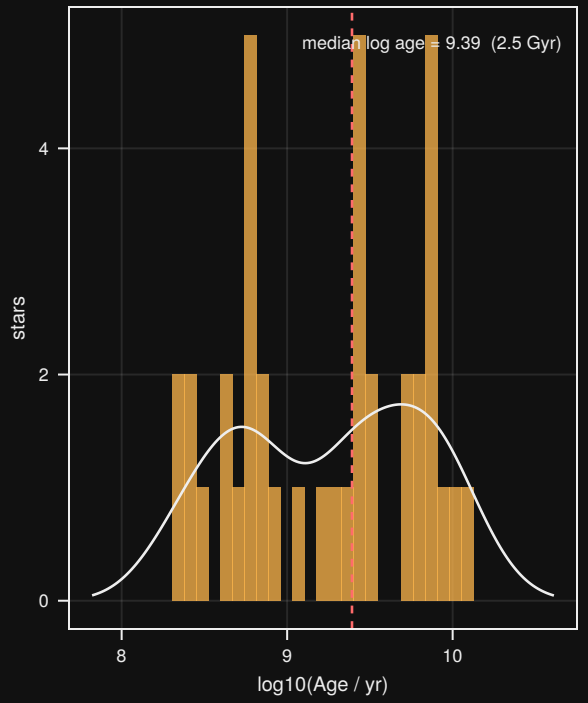
[M/H] GSP-Phot (n = 96)



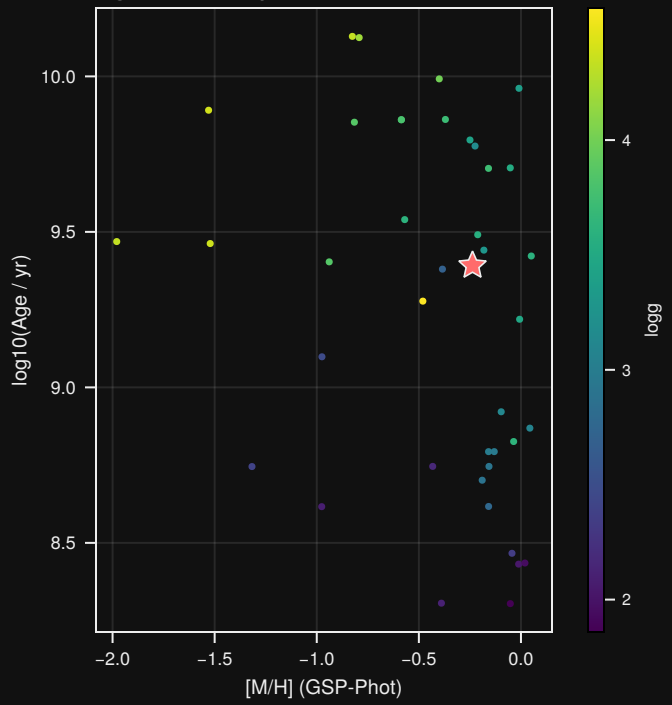
Teff vs [M/H] (n = 96)



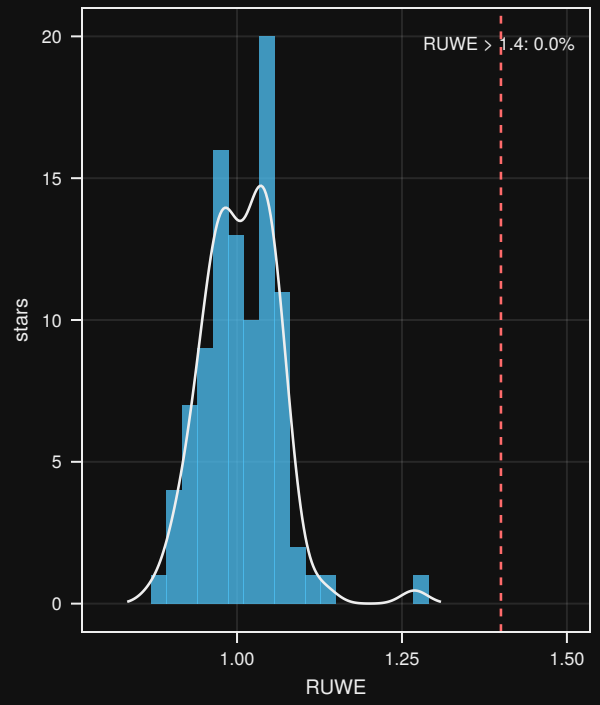
FLAME age (n = 40)



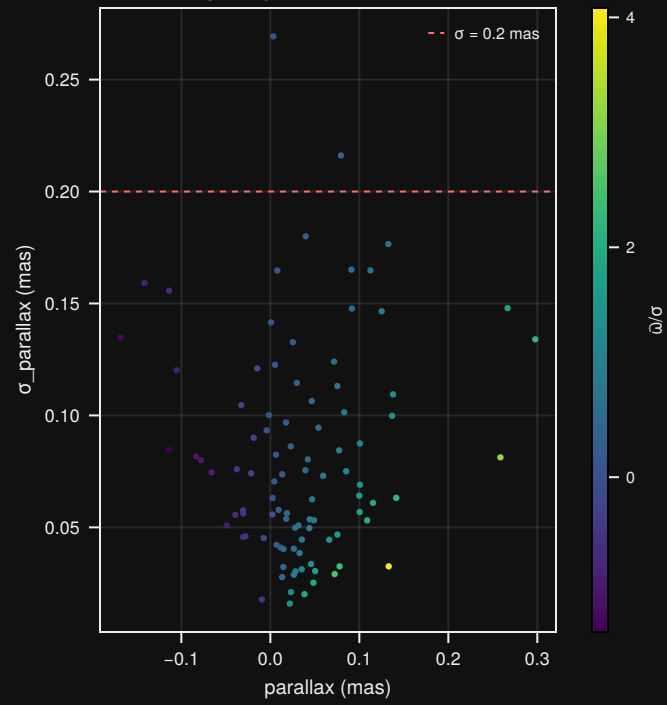
Age–metallicity (n = 40)



RUWE (n = 96)

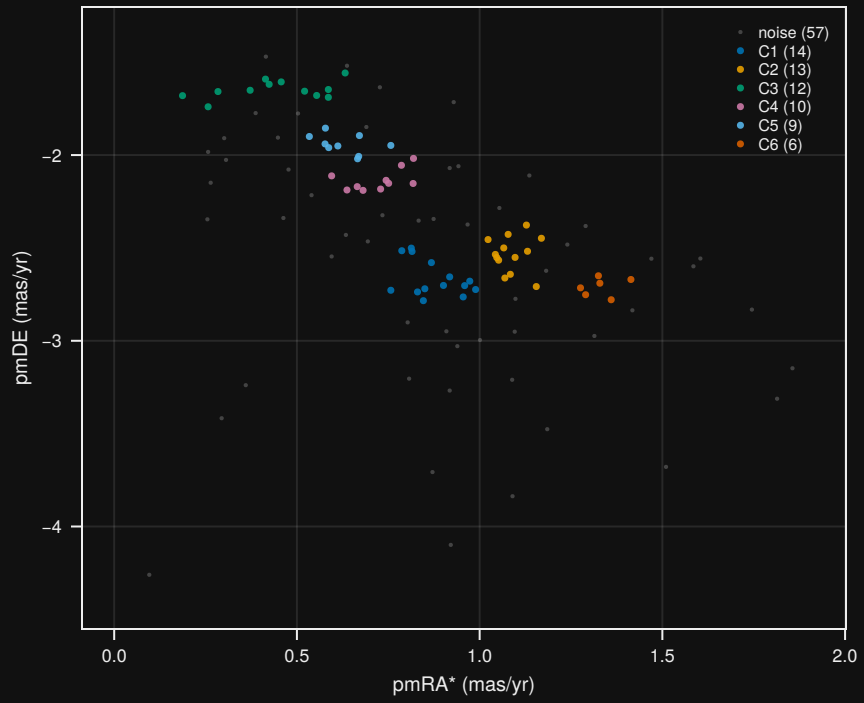


Parallax quality (n = 96)

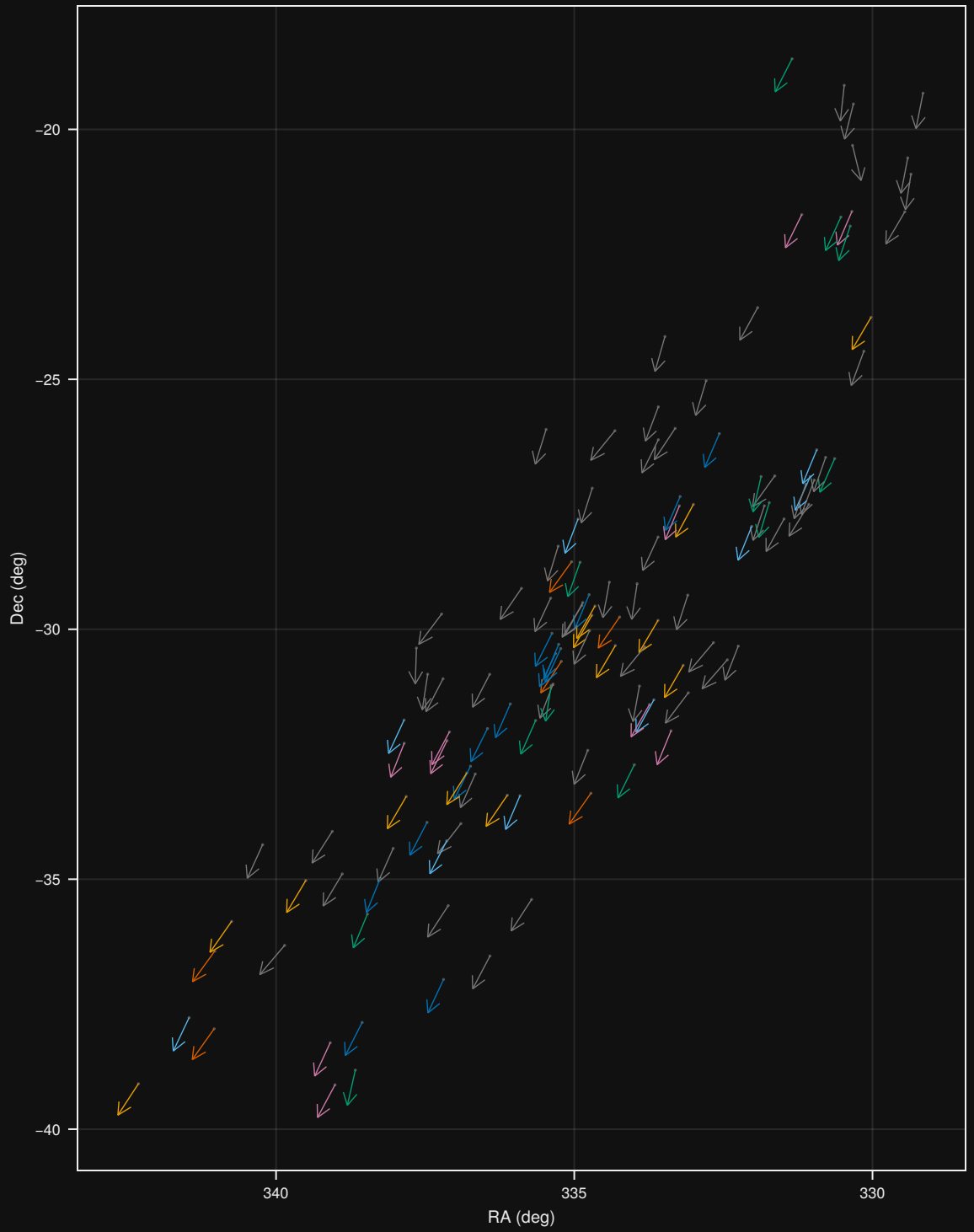


New-25

PM clustering · HDBSCAN · 6 cluster(s) · noise 47%



Proper-motion directions (arrows) coloured by cluster



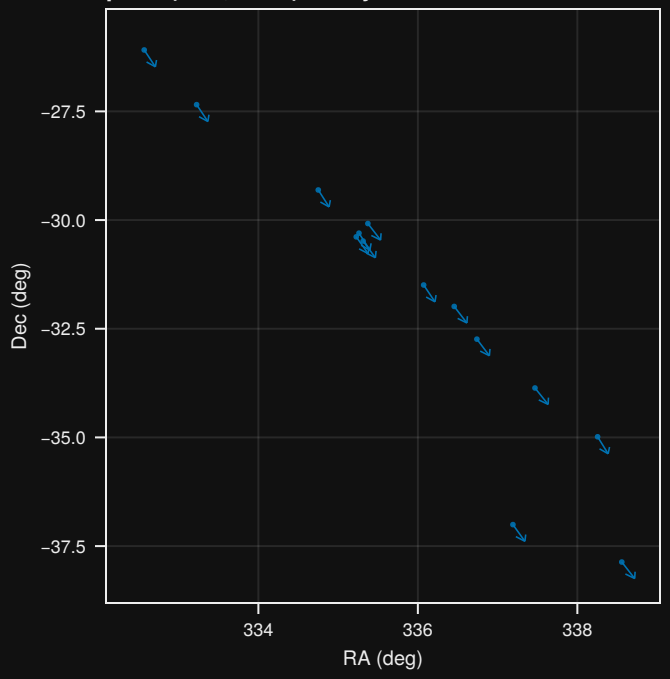
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	57	0.87	-2.48	0.46	-1.02	6
C1	14	0.86	-2.70	0.08	-0.90	2
C2	13	1.08	-2.54	0.05	-1.48	2
C3	12	0.44	-1.65	0.19	-0.40	0
C4	10	0.74	-2.15	0.09	-0.81	0
C5	9	0.61	-1.95	0.08	-1.09	2
C6	6	1.33	-2.70	0.05	-0.77	0

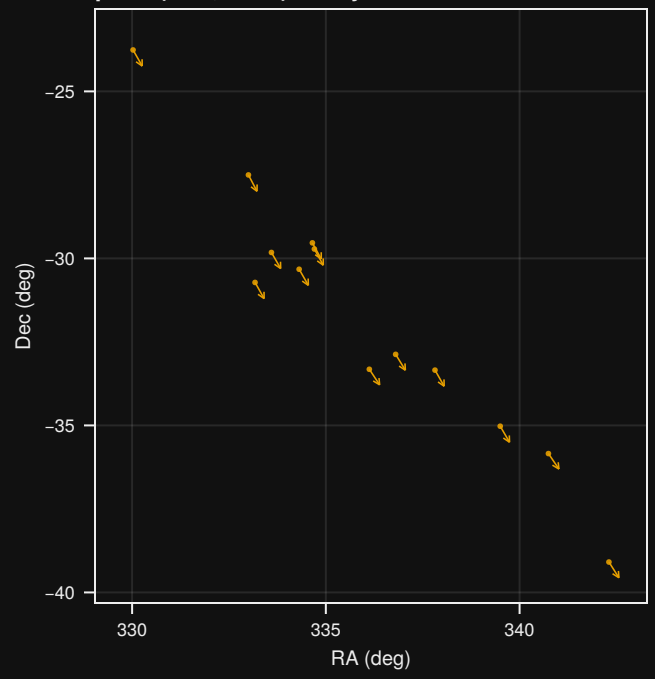
New-25

Per-cluster proper-motion vector maps

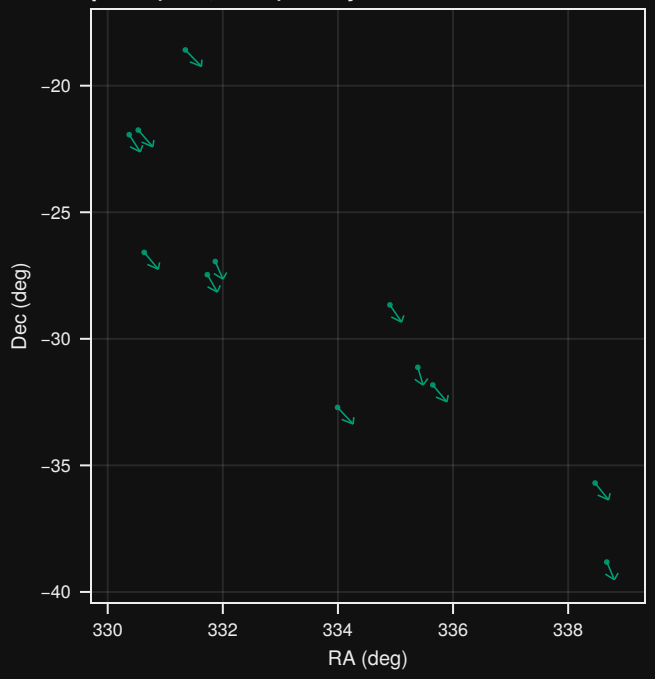
Cluster C1 · n = 14
pm = (0.86, -2.70) mas/yr



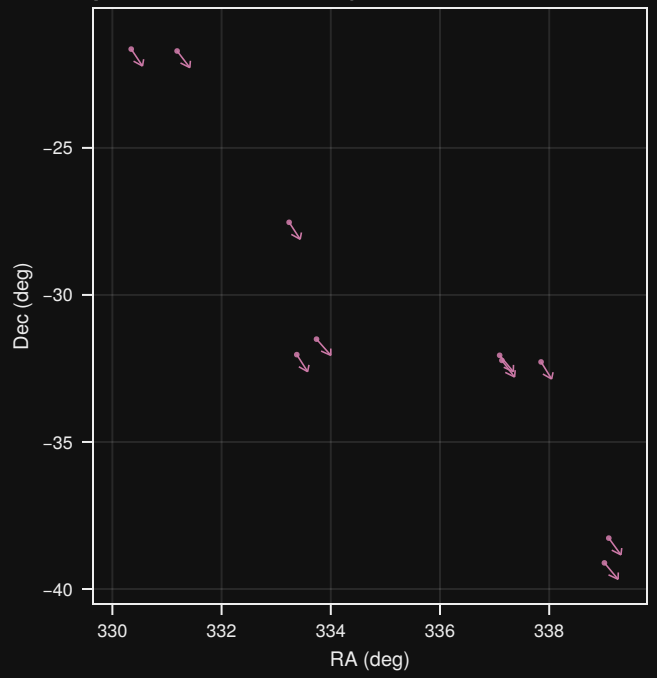
Cluster C2 · n = 13
pm = (1.08, -2.54) mas/yr



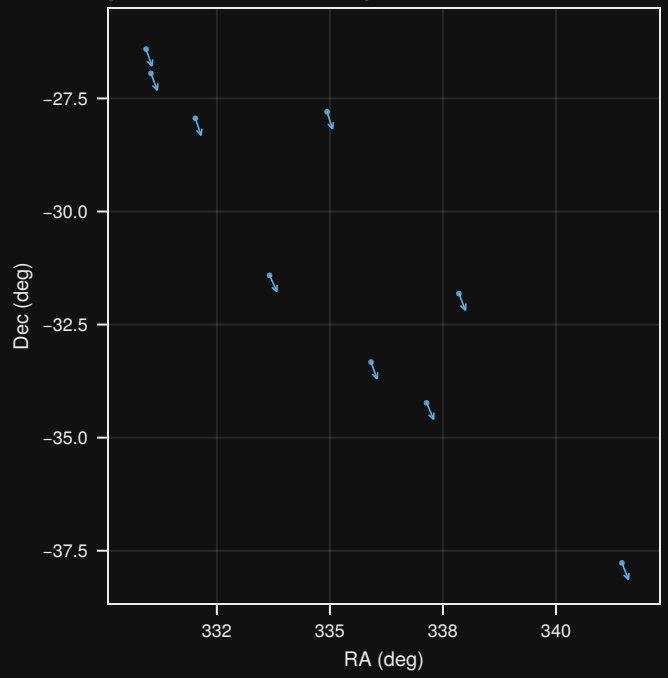
Cluster C3 · n = 12
pm = (0.44, -1.65) mas/yr



Cluster C4 · n = 10
pm = (0.74, -2.15) mas/yr



Cluster C5 · n = 9
pm = (0.61, -1.95) mas/yr



Noise · n = 57

